

Homework 3

Problem 1: One-loop RG for complex scalar

$$\mathcal{L} = \partial_\mu \phi \partial^\mu \phi^* - m^2 |\phi|^2 - \frac{\lambda}{2} (|\phi|^2)^2. \quad (1)$$

Decompose $\phi = (\phi_1 + i\phi_2)/\sqrt{2}$:

$$\mathcal{L} = \frac{1}{2}(\partial\phi_1)^2 + \frac{1}{2}(\partial\phi_2)^2 - \frac{m^2}{2}(\phi_1^2 + \phi_2^2) - \frac{\lambda}{8}(\phi_1^2 + \phi_2^2)^2. \quad (2)$$

Bare Lagrangian in $d = 4 - \epsilon$:

$$\mathcal{L}_B = Z_\phi |\partial\phi|^2 - Z_\phi Z_m m^2 |\phi|^2 - \frac{\lambda}{2} \mu^\epsilon Z_\lambda Z_\phi^2 (|\phi|^2)^2, \quad (3)$$

$Z_i = 1 + \delta_i$. Counterterms:

$$\mathcal{L}_{\text{ct}} = \delta_\phi |\partial\phi|^2 - (\delta_\phi + \delta_m) m^2 |\phi|^2 - \frac{\lambda}{2} \mu^\epsilon (\delta_\lambda + 2\delta_\phi) (|\phi|^2)^2. \quad (4)$$

Self-energy (tadpole $\propto \int d^d k / (k^2 - m^2) = 0$ in dim reg):

$$-i\Sigma(p^2) = \lambda \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2} = \frac{\lambda m^2}{16\pi^2} \left(\frac{2}{\bar{\epsilon}} + 1 - \ln \frac{m^2}{\mu^2} \right), \quad \bar{\epsilon}^{-1} \equiv \epsilon^{-1} - \gamma_E + \ln 4\pi. \quad (5)$$

Hence

$$\delta_\phi = 0, \quad \delta_m = \frac{\lambda}{16\pi^2} \frac{2}{\bar{\epsilon}}, \quad \gamma_m = -\frac{1}{2} \frac{d \ln Z_m}{d \ln \mu} = \frac{\lambda}{16\pi^2}, \quad \gamma_\phi = 0. \quad (6)$$

Four-point function. s -, t -, u -channel bubbles (each symmetry factor 1/2):

$$i\mathcal{M}_{1\text{-loop}} = -i\lambda - i \frac{\lambda^2}{32\pi^2} [I(s) + I(t) + I(u)], \quad I(P^2) = \frac{2}{\bar{\epsilon}} + \text{finite}. \quad (7)$$

At $s = t = u = \mu^2$:

$$\delta_\lambda = \frac{5\lambda}{16\pi^2} \frac{1}{\bar{\epsilon}}. \quad (8)$$

From $\lambda_0 = \mu^\epsilon Z_\lambda \lambda$:

$$\beta(\lambda) = \mu \frac{d\lambda}{d\mu} = -\epsilon\lambda - \lambda\mu \frac{d \ln Z_\lambda}{d \ln \mu} = \frac{5\lambda^2}{16\pi^2} + \mathcal{O}(\lambda^3). \quad (9)$$

$$\boxed{\beta(\lambda) = \frac{5\lambda^2}{16\pi^2}, \quad \gamma_m = \frac{\lambda}{16\pi^2}, \quad \gamma_\phi = 0.} \quad (10)$$

Problem 2: QED counterterms and BPHZ

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\cancel{\partial} - m)\psi + e\bar{\psi}\cancel{A}\psi. \quad (11)$$

Bare:

$$\mathcal{L}_B = -\frac{1}{4}Z_3 F^2 + Z_2 \bar{\psi} i\cancel{\partial} \psi - Z_0 m \bar{\psi} \psi + e\mu^{\epsilon/2} Z_1 \bar{\psi} \cancel{A} \psi. \quad (12)$$

One-loop counterterms ($\overline{\text{MS}}$):

$$\delta_3 = Z_3 - 1 = -\frac{e^2}{12\pi^2} \frac{1}{\epsilon}, \quad (13)$$

$$\delta_2 = Z_2 - 1 = -\frac{e^2}{16\pi^2} \frac{1}{\epsilon}, \quad (14)$$

$$\delta_0 = Z_0 - 1 = -\frac{3e^2}{8\pi^2} \frac{1}{\epsilon}. \quad (15)$$

Ward identity $\Rightarrow Z_1 = Z_2$.

RG equations:

$$\beta(e) = \frac{e^3}{12\pi^2}, \quad (16)$$

$$\gamma_m = \frac{3e^2}{8\pi^2}, \quad (17)$$

$$\gamma_\psi = \frac{e^2}{16\pi^2}, \quad (18)$$

$$\gamma_A = -\frac{e^2}{12\pi^2}. \quad (19)$$

Problem 3: SU(N) Yang–Mills β function

$$\mathcal{L}_{\text{YM}} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + \sum_{f=1}^{N_F} \bar{\psi}_f (i\not{D} - m_f) \psi_f. \quad (20)$$

$g_0 = \mu^{\epsilon/2} Z_g g$. One-loop renormalization constants:

$$Z_3 = 1 - \frac{g^2}{16\pi^2} \left(\frac{5}{3} C_A - \frac{4}{3} T_F N_F \right) \frac{1}{\epsilon}, \quad (21)$$

$$Z_1 = 1 - \frac{g^2}{16\pi^2} \left(\frac{2}{3} C_A + \frac{4}{3} T_F N_F \right) \frac{1}{\epsilon}, \quad (22)$$

$C_A = N$, $T_F = 1/2$. $Z_g = Z_1 Z_3^{-3/2}$:

$$Z_g = 1 + \frac{g^2}{16\pi^2} \left(-\frac{11}{6} N + \frac{1}{3} N_F \right) \frac{1}{\epsilon}. \quad (23)$$

$$\boxed{\beta(g) = -\frac{g^3}{16\pi^2} \left(\frac{11}{3} N - \frac{2}{3} N_F \right)}. \quad (24)$$

Problem 4: Goldstone scattering equivalence

$$\mathcal{L} = \partial_\mu \phi \partial^\mu \phi^* + m^2 |\phi|^2 - \frac{\lambda}{2} (|\phi|^2)^2, \quad v = \frac{m}{\sqrt{\lambda}}. \quad (25)$$

Linear: $\phi = v + (\phi_1 + i\phi_2)/\sqrt{2}$.

$$|\phi|^2 = v^2 + \sqrt{2}v\phi_1 + \frac{\phi_1^2 + \phi_2^2}{2}, \quad -\frac{\lambda}{2} (|\phi|^2)^2 \supset -\frac{\lambda}{8} \phi_2^4 = -\frac{3\lambda}{4!} \phi_2^4. \quad (26)$$

$$i\mathcal{M}_{\phi_2\phi_2 \rightarrow \phi_2\phi_2} = -i(3\lambda). \quad (27)$$

Polar: $\phi = (v + R/\sqrt{2})e^{i\theta/(\sqrt{2}v)}$.

$$|\partial\phi|^2 = \frac{1}{2}(\partial R)^2 + \frac{1}{2}(\partial\theta)^2 + \frac{\sqrt{2}R}{2v}(\partial\theta)^2 + \frac{R^2}{4v^2}(\partial\theta)^2 + \dots \quad (28)$$

Integrating out R (heavy, $m_R^2 = 2\lambda v^2$): s -, t -, u -channel exchange + contact term. Low-

energy limit $s, t, u \ll m_R^2$:

$$i\mathcal{M} = \frac{i}{v^2} \left[\frac{s^2}{s - m_R^2} + \frac{t^2}{t - m_R^2} + \frac{u^2}{u - m_R^2} \right] \xrightarrow{m_R \rightarrow \infty} -\frac{i}{v^2} (s + t + u + 3m_R^2) = -i(3\lambda), \quad (29)$$

using $s + t + u = 0$ (massless Goldstones).

$$\boxed{i\mathcal{M} = -i(3\lambda) \text{ for both parametrizations.}} \quad (30)$$